

RB Weekly AI Brief - Issue 24 - 15.09.2026

Covering the week of 15.09.2026 · Issue 24 of the RB Weekly AI Brief

Recurring themes: Regulatory & HTA Signals (3 of last 4 issues) · Regulation & Policy (3 of last 4 issues) · Healthcare & Life Sciences (3 of last 4 issues) · Models & Research (3 of last 4 issues)

AI News Roundup

Regulatory & HTA Signals

No qualifying HTA news items identified this week. This section requires stories from official HTA body sources or specialist health policy outlets — general AI regulation stories are excluded.

Regulation & Policy

MHRA Commission Publishes Final Recommendations: 'L-Plates' for New AI Models

On 10 September 2026, the National Commission into the Regulation of AI in Healthcare — established by the MHRA in September 2025 and drawing evidence from more than 12,000 clinicians, patients, and industry stakeholders — published its final recommendations for a new UK regulatory framework. This closes the loop from its own "summer 2026" deadline, missed as of early September. The Commission's headline recommendation is that the MHRA introduce staged authorisations for new AI models, likened to 'L-plates' for learner drivers: new systems deployed under close supervision and tight guardrails before graduating to full authorisation based on real-world safety evidence, replacing single point-in-time approval with continuous lifecycle monitoring. The Commission also called for stronger MHRA enforcement powers and a publicly searchable safety database for AI-enabled medical devices.

***So what?** Pharma and health tech companies planning NHS AI product launches must now anticipate a multi-stage regulatory pathway — including supervised deployment periods with post-market performance data requirements — that will extend time-to-full-authorisation and demand robust real-world evidence strategies from the outset; the government's formal response to these recommendations is the next milestone to track.*

MHRA / GOV.UK

Healthcare & Life Sciences

ARPA-H Commits \$62.7M to Autonomous AI Agents for Heart Failure

ARPA-H announced its ADVOCATE program on September 9, 2026, awarding \$33.7 million in first-year contracts to six teams — including Tempus AI, Atman Health, UpDoc, Stanford, Duke, and Kaiser Permanente — to build FDA-authorized autonomous AI agents for heart failure care. The agents are designed to monitor patients continuously, adjust medications, assess symptom severity, and escalate to human clinicians when needed, targeting the roughly 6.7 million Americans with heart failure who struggle to access cardiologists. The program will work directly with the FDA to create a first-ever regulatory framework for autonomous clinical AI.

***So what?** ADVOCATE is actively building the reimbursement pathways and FDA authorization framework for autonomous clinical AI — pharma, HEOR, and market access teams should track this program closely, as its outcomes will shape how agentic AI interventions are evaluated for clinical effectiveness, real-world evidence standards, and payer coverage decisions in chronic disease management.*

MedTech Dive

Models & Research

OpenAI Releases GPT-6 Astra

OpenAI released GPT-6 Astra on September 3, 2026, its largest training run to date at over 100,000 GPUs on its Stargate facility in Texas, with earlier OpenAI models playing a role in supervising the new model's training for the first time. Astra is the first OpenAI model to cross the 'Critical' cybersecurity threshold under its Preparedness Framework, prompting a phased rollout gated behind its Daybreak program before wider availability. On headline benchmarks OpenAI reports strong scores — including 99.9% on ARC-AGI-3 and 97.6-98% on FrontierMath Tier 4 — though both carry real caveats: the ARC-AGI-3 figure depends on OpenAI's own stateful evaluation harness (independent stateless runs score substantially lower), and Epoch AI, which runs FrontierMath, discloses that OpenAI funded the benchmark's development and holds exclusive access to part of it. Astra is not unambiguously the strongest model available: on the one index OpenAI itself presents as cross-vendor comparable, Astra trails Anthropic's Claude Fable 5.1, and Meta's Muse Spark 1.3 outperforms it on coding benchmarks.

***So what?** For pharma and HEOR teams, Astra's genuine advances — particularly in scientific reasoning and multi-step autonomous research workflows — are worth evaluating for evidence synthesis and literature review use cases, but headline benchmark claims should be read against the harness and funding caveats above rather than taken at face value, the same discipline worth applying to any vendor's self-reported 'state of the art' claim.*

OpenAI

DeepSeek Releases V4.1-Flash, Outperforming Its Own Flagship Model

DeepSeek released DeepSeek-V4.1-Flash on September 10, 2026 — a 552-billion-parameter open-weight mixture-of-experts model with native multimodal vision and a 1-million-token context window — and reported it comprehensively surpasses the larger V4-Pro on performance, speed, and cost. Benchmark scores include 90.6 on Terminal-Bench 2.1 (narrowly above Claude Opus 5's 89.1), 74.2% on DeepSWE v1.1, and 90.9 on GPQA Diamond — though it trails Opus 5 on the latter (93.4). DeepSeek immediately began routing V4-Pro API calls to V4.1-Flash at the smaller model's rates, effectively making frontier-level performance available at near-commodity cost.

***So what?** The availability of open-weight, frontier-grade reasoning models at very low cost lowers the barrier for pharma and HEOR teams to deploy AI at scale for tasks such as real-world data analysis, dossier preparation, and health technology assessment submissions — but it also raises urgent questions about model validation, data governance, and auditability that life sciences compliance functions will need to address.*

SiliconANGLE

Academic Paper Summaries

Selected from PubMed · Published within the last 12 months · New selections each week

Domain Paper — HEOR / Health Economics / Market Access

Optimizing deep learning-enhanced opportunistic osteoporosis screening from chest X-rays in Australia: Impact of follow-up and treatment adherence on clinical and economic outcomes.

Reginster JY, Ebeling PR, Alokail M, Jung J, Hilgsmann M. · Bone · 2026

#HTA · #MarketAccess · #RealWorldEvidence

This Australian microsimulation Markov model evaluated AI-enhanced opportunistic osteoporosis screening from routine chest X-rays against no screening in women aged 50+. The AI-enhanced approach was a dominant strategy — better health outcomes (+0.0025 QALYs, 43 fractures prevented per 10,000 women screened) at lower lifetime cost (AUD590,000 in savings per 10,000 screened) — a rare combination in health economics where interventions usually trade one against the other. For HEOR and market access teams, this is genuine formal cost-effectiveness modelling (Markov microsimulation, QALYs, dominant-strategy analysis) rather than a diagnostic-accuracy study, and it flags follow-up and adherence as the real economic lever, not the AI detection step alone.

PMID: 42732841	PubMed →	DOI →
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AI Research Paper 1

Large Language Models in Neurological Practice: Real-World Study.

Maiorana NV, Marceglia S, Treddenti M, Tosi M, Guidetti M, Creta MF, Bocci T, Oliveri S, Martinelli Boneschi F, Priori A. · Journal of medical Internet research · 2025

#ClinicalAI · #Diagnostics · #PatientOutcomes

This study tested whether off-the-shelf AI chatbots (ChatGPT and Gemini) could match real neurologists in diagnosing patients from actual clinical records at an Italian hospital. Neurologists outperformed both AI tools, achieving 75% diagnostic accuracy compared to lower rates for the AI models, which also required additional prompting to reach useful answers. The findings suggest that general-purpose AI tools are not yet ready to replace specialist clinical judgment, though they may still play a supporting role in neurology workflows.

PMID: 40982758	PubMed →	DOI →
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AI Research Paper 2

Artificial intelligence coupled to pharmacometrics modelling to tailor malaria and tuberculosis treatment in Africa.

Turon G, Mulubwa M, Montaner A, Njoroge M, Chibale K, Duran-Frigola M. · Nature communications · 2025

#Genomics · #DrugDevelopment · #RealWorldEvidence

This study developed an AI-driven pipeline to identify which genetic variants in African populations could affect how patients respond to malaria and tuberculosis drugs, and to recommend personalized dose adjustments. By combining machine learning with established pharmacokinetic modeling, the approach addresses a longstanding gap caused by drug development historically being conducted outside Africa. The work has direct implications for improving treatment outcomes in underserved populations and advancing pharmacogenomics for infectious diseases in Africa.

PMID: 41115876	PubMed →	DOI →
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